CV of Santu Karmaker

Email: santu@ucf.edu
Portfolio: https://karmake2.github.io/ Mobile: +1-217-979-3244
Google Scholar: https://scholar.google.com/citations?user=y6pZKT4AAAAJ

Academic Appointments

- **University of Central Florida** — Orlando, Florida, US
  Assistant Professor, Department of Computer Science — *August 2024 - Current*
- **Auburn University** — Auburn, Alabama, US
  Assistant Professor, Department of Computer Science and Software Engineering — *Jan 2020 - July 2024*

Research Focus Areas

- Natural Language Processing (**NLP**), Information Retrieval (**IR**), Democratization of AI/ML
- According to CSRankings.org, **one of the most prolific assistant professors of NLP in Florida**

Academic Preparation

- **Massachusetts Institute of Technology (MIT)** — Boston, Massachusetts, US
  Postdoctoral Research Associate — *Jan 2019 - Dec 2019*
  Laboratory for Information & Decision Systems (LIDS)
  Host: *Dr. Kalyan Veeramachaneni*
- **University of Illinois at Urbana-Champaign (UIUC)** — Urbana, Illinois, US
  Ph.D. in Computer Science — *Aug 2014 - May 2019*
  Advisor: *Prof. ChengXiang Zhai*
  Thesis Committee Members: Prof. Jiawei Han, Prof. Hari Sundaram, Dr. Hao Ma
- **Bangladesh University of Engineering and Technology (BUET)** — Dhaka, Bangladesh
  M.S. in Computer Science & Engineering, CGPA: ***4.00*** *out of 4.00.* ***[Rank: 1]*** — *Apr 2012 - Apr 2014*
- **Bangladesh University of Engineering and Technology (BUET)** — Dhaka, Bangladesh
  B.S. in Computer Science & Engineering, CGPA: ***3.98*** *out of 4.00.* ***[Rank: 1]*** — *Jun 2007 - Feb 2012*

Grants and Awards (≈**$2.2 million**)

- [**ONR Young Investigator Program (YIP) Awarded $750,000 as Single PI**] **(2026-29)** Basic Research Grant. Program - Warfighter Performance (Code 34), Topic - Naval Strike Group Training and Assessment. Sponsor: Office of Naval Research (**ONR**). Project Title: "Rapid Holistic Summarization for Decision Superiority in Urgent Scenarios". Award Link
- [**AFOSR Awarded $542,485 as Single PI**] **(2023-26)** Basic Research Grant. Program - Information & Networks, Topic - Trust & Influence. Sponsor: Air Force Office of Scientific Research (**AFOSR**). Title: "A Novel Human-AI Collaborative Framework For Multi-Perspective Narrative Analytics & Braiding At Scale". Award #: FA95502310426. Award Link
- [**NSF Awarded $700,854 as Lead PI (2023-25)**] Program: Research on Emerging Technologies for Teaching and Learning (RETTL). Sponsor: National Science Foundation (**NSF**).
  Title: "An Intelligent Assistant to Support Teachers and Students in Simulation-Based Science Learning". Award #: 2302974. Award Link
- [**ARO Awarded $60,000 as Single PI (2022-23)**] Short-Term Innovative Research (STIR) Grant. Sponsor: Army Research Office (**ARO**). Title: "Semantic Machine For Robust Interpretation Of Noisy Intelligence". Award #: W911NF2210280. Award Link
- [**USDA Awarded $50,000 as Lead PI (2022)**] Champion in "Food for Thought" NLP challenge hosted by Coleridge Initiative in collaboration with **USDA**. News 1 News 2.
- [**ASU Awarded $23,135 as Single PI (2021)**] Deep Time Series Forecasting Techniques for modeling Molecular Dynamics. Sponsor: Arizona State University. Title: "What Does 'Self' Look Like?"

Industry Experience

- **Microsoft Research (Mentor: Riham Mansour)** — Bellevue, Seattle, US
  Summer Research Intern @FUSE labs — *Summer, 2018*
  Project: Developed semantic representations for understanding utterances for LUIS
- **Microsoft Research (Mentor: Hao Ma)** — Redmond, Seattle, US
  Summer Research Intern @Internet Service Research Center (ISRC) — *Summer, 2017*
  Project: Automatic Self-Evolving Text Generation
- **Yahoo Research (Mentor: Yi Chang)** — Sunnyvale, California, US
  Summer Research Intern @Search Science Group — *Summer, 2016*
  Project: Influence Modeling for User Search Behavior
- **@WalmartLabs (Mentor: Parikshit Sondhi)** — Sunnyvale, California, US
  Summer Research Intern @Search Relevance Team — *Summer, 2015*
  Project: Learning-to-Rank for E-Commerce Search
- **Jump Trading LLC (Mentor: Qiaozhu Mei)** — Chicago, Illinois, US
  Text Mining Intern @Search Relevance Team — *Fall 2016 - Spring 2017*
  Project: *Confidential and Restricted*

Other Awards

- Recipient of **DAAD (German Academic Exchange Service) Faculty Fellowship 2024.**
- **100+ Women Strong Leadership in Diversity Faculty/Staff Award 2024** at the Samuel Ginn College of Engineering at Auburn.
- **Auburn University Undergraduate Research Mentor Award, 2023**
- **Best Poster Award Nomination at CIKM 2020**

Peer-Reviewed Journal Articles (My students are underlined)

- [**TACL 2026**]: R. Alexander Knipper, Charles S. Knipper, Kaiqi Zhang, Valerie Sims, Clint Bowers, **Santu Karmaker**. *"The Bias is in the Details: An Assessment of Cognitive Bias in LLMs"*. Accepted in the Transactions of the Association for Computational Linguistics (TACL), 2026. **(Impact Factor: 11.7).**
- [**TMLR 2024**]: Md. Mahadi Hassan, Alex Knipper and **Shubhra K. Karmaker Santu**. *"Introducing 'Forecast Utterance' for Conversational Data Science"*. In Transactions on Machine Learning Research, 2024.
- [**IP&M 2023**]: D. Feng and **Shubhra Kanti Karmaker Santu**. *"Joint Upper & Expected value Normalization for Evaluation of Retrieval Systems: A Case Study with Learning-to-Rank methods"*. Information Processing & Management 60.4 (2023): 103404. **(Impact Factor: 8.6).**
- [**TIST 2023**]: S. Sarkar, B. S. Bijoy, S. J. Saba, D. Feng, Y. Mahajan, S. R. Islam, Md. R. Amin and **Shubhra Kanti Karmaker Santu**. *"Ad-Hoc Monitoring of COVID-19 Global Research Trends for Well-Informed Policy Making"*. ACM Transactions on Intelligent Systems and Technology, 14(2), pp.1-28. **(Impact Factor: 10.7)**.
- [**CSUR 2022**]: **Shubhra Kanti Karmaker Santu**, Md. Mahadi Hassan, Micah J. Smith, Lei Xu, ChengXiang Zhai, Kalyan Veeramachaneni. "*AutoML to Date and Beyond: Challenges and Opportunities*". ACM Computing Surveys (CSUR), 54(8), pp.1-36. **(Impact Factor: 30.4)**.
- [**SIGKDD Explorations 2018**]: **Shubhra Kanti Karmaker Santu**, C. Geigle, D. C. Ferguson, W. Cope, M. Kalantzis, D. Searsmith, Chengxiang Zhai. "SOFSAT: Towards a Setlike Operator based Framework for Semantic Analysis of Text". ACM SIGKDD Explorations Newsletter, 20(2), pp.21-30.

Peer-Reviewed Conference Articles (My students are underlined)

- [**NeurIPS 2026**]: Eftekhar Hossain, John Salvador, **Santu Karmaker**. "Evaluating Whether LLMs Can Reliably Connect the DOTs?" In Proceedings of the Fortieth Annual Conference on Neural Information Processing Systems: Evaluations & Datasets Track. **(Acceptance Rate: $971/3,757 = 25.8\%$).**
- [**NeurIPS 2026**]: Adam Bawatneh, Sagar Sapkota, Amrit Singh Bedi, **Santu Karmaker**, Mubarak Shah. "OmniToM: Benchmarking Theory of Mind in LLMs via Explicit Belief Modeling." In Proceedings of the Fortieth Annual Conference on Neural Information Processing Systems: Evaluations & Datasets Track. **(Acceptance Rate: $971/3,757 = 25.8\%$).**
- [**EMNLP 2026**]: Faria Binte Kader, Mohtasim Hadi Rafi, Shah Wasif Sazzad, **Santu Karmaker**. "Exploring LLMs for South Asian Music Understanding and Generation". To appear in Findings of the 2026 Conference on Empirical Methods in Natural Language Processing. **(Acceptance Rate: $2,533/17,669 = 14.3\%$).**

- [**ACL 2026**]: Eshgin Hasanov, Md. Mahadi Hassan, **Santu Karmaker**, Aashish Yadavally. "The Path Not Taken: Duality in Reasoning about Program Execution." In Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pages 16165–16180, San Diego, California, United States. Association for Computational Linguistics. **(Acceptance Rate: $2,296/12,148 = 18.9\%$).**
- [**ACL 2026**]: Md. Mahadi Hassan, John Salvador, Akond Ashfaque Ur Rahman, **Santu Karmaker**. "Large Language Models for IT Automation Tasks: Are We There Yet?" In Findings of the Association for Computational Linguistics: ACL 2026, pages 11534–11573, San Diego, California, United States. Association for Computational Linguistics. **(Acceptance Rate: $2,163/12,148 = 17.8\%$).**
- [**AACL 2026**]: Yash Mahajan, Naman Bansal, Faria Binte Kader, **Santu Karmaker**. "Beyond Downstream Task Performances: A Set-Theoretic Evaluation of Sentence Embeddings." In Proceedings of the 5th Asia-Pacific Chapter of the Association for Computational Linguistics & the 15th International Joint Conference on Natural Language Processing. Association for Computational Linguistics. **(Acceptance Rate: Not published yet).**
- [**AACL 2026**]: Md. Mahadi Hassan, Sazzadur Rahaman, Yusuf Sarwar Uddin, **Santu Karmaker**. "Private Evaluation Protocol: Contamination-Resistant Black-Box Evaluation via Public-Slice Surrogates." In Proceedings of the 5th Asia-Pacific Chapter of the Association for Computational Linguistics & the 15th International Joint Conference on Natural Language Processing. Association for Computational Linguistics. **(Acceptance Rate: Not published yet).**
- [**EMNLP 2025**]: John Salvador, Naman Bansal, Mousumi Akter, Souvika Sarkar, Anupam Das, **Santu Karmaker**. "Benchmarking LLMs on the Semantic Overlap Summarization Task." In Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing, pages 33352–33373, Suzhou, China. Association for Computational Linguistics. **(Acceptance Rate: $1811/8174 = 22.16\%$).**
- [**ACL 2025**]: Mohammad Saqib Hasan, Saikat Chakraborty, **Santu Karmaker**, Niranjan Balasubramanian. "Teaching an Old LLM Secure Coding: Localized Preference Optimization on Distilled Preferences." In Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pages 26039–26057, Vienna, Austria. Association for Computational Linguistics. **(Acceptance Rate: $1,699/8,360 = 20.3\%$).**
- [**NAACL 2025**]: Eftekhar Hossain, Sanjeev Kumar Sinha, Naman Bansal, Alex Knipper, Souvika Sarkar, John Salvador, Yash Mahajan, Sri Guttikonda, Mousumi Akter, Md. Mahadi Hassan, Matthew Freestone, Matthew C Williams Jr, Dongji Feng, **Santu Karmaker**. "LLMs as Meta-Reviewers' Assistants: A Case Study". In Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers), pages 7763–7803, Albuquerque, New Mexico. Association for Computational Linguistics. **(Acceptance Rate: $719/3185 = 22\%$).**
- [**AACL 2025**]: Yash Mahajan, Matthew Freestone, Naman Bansal, Sathyanarayanan N. Aakur, **Santu Karmaker**. "Revisiting Word Embeddings in the LLM Era." In Proceedings of the 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics, pages 2686–2717, Mumbai, India. The Asian Federation of Natural Language Processing and The Association for Computational Linguistics. **(Acceptance Rate: $232/1456 = 19\%$.)**
- [**EMNLP 2024**]: Yash Mahajan, Naman Bansal, Eduardo Blanco, **Santu Karmaker**. "ALIGN-SIM: A Task-Free Test Bed for Evaluating and Interpreting Sentence Embeddings through Semantic Similarity Alignment." In Findings of the Association for Computational Linguistics: EMNLP 2024, pages 7393–7428, Miami, Florida, USA. **(Acceptance Rate: $1029/6105 = 16.9\%$).**
- [**FSE 2024**]: Md. Mahadi Hassan, John Salvador, **Shubhra Kanti Karmaker Santu**, Akond Rahman. "State Reconciliation Defects in Infrastructure as Code". In ACM International Conference on the Foundations of Software Engineering (FSE), 2024: 1865-1888. **(Acceptance Rate: $121/474 = 25.5\%$).**
- [**Learning@Scale 2024**]: Effat Farhana, Souvika Sarkar, Ralph Knipper, Indrani Dey, Hari Narayanan, Sadhana Puntambekar, **Santu Karmaker**. "SimPal: Towards a Meta-Conversational Framework to Understand Teacher's Instructional Goals for K-12 Physics". Accepted for publication at L@S '24: Proceedings of the Eleventh ACM Conference on Learning @ Scale Proceedings: 461-465
- [**ICPE 2024**]: Souvika Sarkar, Mohammad Fakhruddin Babar, Md. Mahadi Hassan, Monowar Hasan, **Shubhra Kanti Karmaker Santu**. "Processing Natural Language on Embedded Devices: How Well Do Modern Models Perform?" In International Conference on Performance Engineering (ICPE), 2024: 211-222. **(Acceptance Rate: $12/52 = 23.07\%$).**
- [**KONVENS 2024**]: Mousumi Akter, **Shubhra Kanti Karmaker**. "Redundancy Aware Multiple Reference Based Gainwise Evaluation of Extractive Summarization." In Proceedings of the 20th Conference on Natural Language Processing (KONVENS 2024): 182-195.
- [**EMNLP 2023**]: Souvika Sarkar, Dongji Feng and **Shubhra Kanti Karmaker Santu**. *"Zero-Shot Multi-Label Topic Inference with Sentence Encoders"*. In Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing: 16218-16233. **(Acceptance Rate: $1047/4909 = 21.3\%$).**
- [**EMNLP 2023**]: Mousumi Akter, Souvika Sarkar and **Shubhra Kanti Karmaker Santu**. *"On Evaluation of Bangla Word Analogies"*. In Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing: 13121-13127. **(Acceptance Rate: $1047/4909 = 21.3\%$).**

- [**EMNLP 2023**]: **Shubhra Kanti Karmaker Santu** and Dongji Feng. *"TELeR: A General Taxonomy of LLM Prompts for Benchmarking Complex Tasks"*. In Findings of the 2023 Conference on Empirical Methods in Natural Language Processing: 14197-14203.**(Acceptance Rate: 1060/3862 = 27.4%).**
- [**ACL 2022**]: M. Akter, N. Bansal, **Shubhra Kanti Karmaker Santu**. *"Revisiting Automatic Evaluation of Extractive Summarization Task: Can We Do Better than ROUGE?"* In Findings of the Association for Computational Linguistics: ACL 2022 (pp. 1547-1560). **(Acceptance Rate: 361/2677 = 13.49%).**
- [**EMNLP 2022**]: N. Bansal, M. Akter, **Shubhra Kanti Karmaker Santu**. *"Learning to Generate Overlap Summaries through Noisy Synthetic Data"*. In Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (pp. 11765-11777). **(Acceptance Rate: 715/3242 = 22.1%)**
- [**EMNLP 2022**]: N. Bansal, M. Akter, **Shubhra Kanti Karmaker Santu**. *"SEM-F1: an Automated Metric for Evaluating Multi-Narrative Overlap Summaries"*. In Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (pp. 780-792). **(Acceptance Rate: 715/3242 = 22.1%).**
- [**COLING 2022**]: N. Bansal, M. Akter, **Shubhra Kanti Karmaker Santu**. *"Semantic Overlap Summarization among Multiple Alternative Narratives: an Exploratory Study"*. In Proceedings of the 29th International Conference on Computational Linguistics (pp. 6195-6207). **(Acceptance Rate: 522/1563 = 33.4%).**
- [**AACL 2022**]: R. Knipper, Md. Mahadi Hassan, Mehdi Sadi and **Shubhra Kanti Karmaker**. *"Analogy-Guided Evolutionary Pretraining of Binary Word Embeddings"*. In Proceedings of the 2nd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 12th International Joint Conference on Natural Language Processing (Volume 1: Long Papers) (pp. 683-693). **(Acceptance Rate: 147/554 = 26.5%).**
- [**AACL 2022**]: Souvika Sarkar, Dongji Feng and **Shubhra Kanti Karmaker**. *"Exploring Universal Sentence Encoders for Zero-shot Text Classification"*. In Proceedings of the 2nd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 12th International Joint Conference on Natural Language Processing (pp. 135-147). **(Acceptance Rate: 147/554 = 26.5%).**
- [**BigData 2022**] : Y. Mahajan, S. R. Islam, Md. R. Amin, **Shubhra K Karmaker Santu**. "*Data-Driven Estimation of Effectiveness of COVID-19 Non-pharmaceutical Intervention Policies*". In 2022 IEEE International Conference on Big Data (Big Data) (pp. 5312-5321). IEEE. **(Acceptance Rate: 19.2%).**
- [**IUI 2021**] : Biddut S. Bijoy, Syeda J. Saba, Souvika Sarkar, Md Saiful Islam, Sheikh R. Islam, Md. Ruhul Amin, **Shubhra K Karmaker Santu**. "*COVID19α: Interactive Spatio-Temporal Visualization of COVID-19 Symptoms through Tweet Analysis*". In 26th International Conference on Intelligent User Interfaces-Companion (pp. 28-30). **(Acceptance Rate: 12/34 = 35.29%).**
- [**CIKM 2020 (**Best Poster Nomination**)**] : **Shubhra Kanti Karmaker Santu**, Parikshit Sondhi and ChengXiang Zhai. "*Empirical Analysis of Impact of Query-Specific Customization of nDCG: A Case-Study with Learning-to-Rank Methods*". In Proceedings of the 29th ACM International Conference on Information & Knowledge Management (pp. 3281-3284). **(Acceptance Rate: 60/161 = 37.26%).**
- [**ICWSM 2020**]: Naeemul Hassan, Amrit Poudel, Jason Hale, Claire Hubacek, Khandakar Tasnim Huq, **Shubhra Kanti Karmaker Santu**, Syed Ishtiaque Ahmed. "*Towards Automated Sexual Violence Report Tracking*". In Proceedings of the International AAAI Conference on Web and Social Media (Vol. 14, pp. 250-259). **(Acceptance Rate: 72/295 = 24.4%).**
- [**CoNLL 2019**]: **Shubhra Kanti Karmaker Santu**, Kalyan Veeramachaneni, ChengXiang Zhai. "Neural Language Models with Evolving Topical Influence". In Proceedings of the 23rd Conference on Computational Natural Language Learning (CoNLL) (pp. 778-788). **(Acceptance Rate: 97/441 = 22.0%).**
- [**CIKM 2019**]: Saar Kuzi, Sahiti Labhishetty, **Shubhra Kanti Karmaker Santu**, Prasad Pradip Joshi and ChengXiang Zhai. "Analysis of Adaptive Training for Learning to Rank in Information Retrieval". In Proceedings of the 28th ACM International Conference on Information and Knowledge Management (pp. 2325-2328). **(Acceptance Rate: 100/470 = 21.3%).**
- [**CIKM 2018**]: **Shubhra Kanti Karmaker Santu**, Liangda Li, Yi Chang, ChengXiang Zhai. "JIM: Joint Influence Modeling for Collective Search Behavior". In Proceedings of the 27th ACM International Conference on Information and Knowledge Management (pp. 637-646). **(Acceptance Rate: 147/862 = 17.0%)**
- [**CIKM 2017**]: Yiren Wang, Dominic Seyler, **Shubhra Kanti Karmaker Santu**, ChengXiang Zhai. "A Study of Feature Construction for Text-based Forecasting of Time Series Variables". In Proceedings of the 2017 ACM Conference on Information and Knowledge Management (pp 2347-2350). **(Acceptance Rate: 119/419 = 28.4%)**
- [**SIGIR 2017**]: **Shubhra Kanti Karmaker Santu**, Parikshit Sondhi, ChengXiang Zhai . "On Application of Learning to Rank for E-Commerce Search". In Proceedings of the 40th international ACM SIGIR conference on research and development in information retrieval (pp. 475-484). **(Acceptance Rate: 78/362 = 22.0%).**
- [**WWW 2017**]: **Shubhra Kanti Karmaker Santu**, Liangda Li, Dae Hoon Park, Yi Chang, ChengXiang Zhai. "Modeling the Influence of Popular Trending Events on User Search Behavior". In Proceedings of the 26th International Conference on World Wide Web Companion (pp. 535-544). **(Acceptance Rate: 200/966 = 20.7%)**

- [**CIKM 2016**]: **Shubhra Kanti Karmaker Santu**, Parikshit Sondhi, ChengXiang Zhai . "Generative Feature Language Models for Mining Implicit Features from Customer Reviews". In Proceedings of the 25th ACM International on conference on Information and knowledge management (pp. 929-938). **(Acceptance Rate: 160/701 = 23.0%).**
- [**IJCNN 2014**]: Md. Mustafizur Rahman, **Shubhra Kanti Karmaker Santu**, Md. Monirul Islam, Kazuyuki Murase. "Forecasting time series - A layered ensemble architecture". In 2014 International Joint Conference on Neural Networks (IJCNN) (pp. 210-217). IEEE.
- [**CEC 2014**]: **Shubhra Kanti Karmaker Santu**, Md. Mustafizur Rahman, Md. Monirul Islam, Kazuyuki Murase. "Towards better generalization in Pittsburgh learning classifier systems". In 2014 IEEE Congress on Evolutionary Computation (CEC) (pp. 1666-1673). IEEE.

Peer-Reviewed Workshop Articles (My students are underlined)

- [**CMN 2026**]: Hoang Bao Truong, Inderjeet Mani, Parisa Kordjamshidi and **Santu Karmaker**. *"Natural Language versus Narrative Representation in Causal Video Question Answering."* In Proceedings of 9TH International Workshop on Computational Models of Narrative, (CMN'26), Universidad Complutense de Madrid, Spain.
- [**LoResLM@EACL 2026**]: Abdullah Khan Zehady, Shubhashis Roy Dipta, Naymul Islam, Safi Al Mamun, and **Santu Karmaker**. *"BanglaLlama: LLaMA for Bangla Language."* In Proceedings of the Second Workshop on Language Models for Low-Resource Languages (LoResLM 2026), pages 73–89, Rabat, Morocco. Association for Computational Linguistics.
- [**BLP@AACL 2025**]: Nishat Raihan, Mohammad Anas Jawad, Md Mezbaur Rahman, Noshin Ulfat, Pranav Gupta, Mehrab Mustafy Rahman, **Santu Karmaker**, and Marcos Zampieri. *"Overview of BLP-2025 Task 2: Code Generation in Bangla"*. In Proceedings of the Second Workshop on Bangla Language Processing (BLP-2025), pages 373–387, Mumbai, India. Association for Computational Linguistics.
- [**BLP@AACL 2025**]: Souvika Sarkar, Md Najib Hasan, and **Santu Karmaker**. *"Zero-Shot Multi-Label Classification of Bangla Documents: Large Decoders Vs. Classic Encoders"*. In Proceedings of the Second Workshop on Bangla Language Processing (BLP-2025), pages 91–106, Mumbai, India. Association for Computational Linguistics.
- [**UserNLP@WWW 2022**]: Souvika Sarkar, **Shubhra Kanti Karmaker Santu**. *"Concept Annotation from Users Perspective: A New Challenge"*. In UserNLP: User-centered Natural Language Processing Workshop @Companion Proceedings of the Web Conference 2022 (pp. 1180-1188).
- [**WPES 2018**]: **Shubhra Kanti Karmaker Santu**, Vincent Bindschaedler, ChengXiang Zhai, Carl A. Gunter. "NRF: A Naive Re-identification Framework". In Proceedings of the 2018 Workshop on Privacy in the Electronic Society (pp. 121-132).

Pre-Prints and Technical Reports (My students are underlined)

- [**Arxiv 2023**]: Mousumi Akter and **Shubhra Kanti Karmaker Santu**. *"FaNS: a Facet-based Narrative Similarity Metric"*. CoRR abs/2309.04823 (2023).
- [**Arxiv 2023**]: Md. Mahadi Hassan, Alex Knipper and **Shubhra Kanti Karmaker Santu**. *"ChatGPT as your Personal Data Scientist"*. CoRR abs/2305.13657 (2023).
- [**RWDS 2023**]: Alex Knipper, Naman Bansal, Jingyi Zheng, Wenying Li, and **Shubhra Kanti Karmaker Santu**. *"Food for Thought: First place winners – Auburn Big Data"*. Real World Data Science, August 21, 2023.
- [**Arxiv 2019**]: Lei Xu, **Shubhra Kanti Karmaker Santu**, Kalyan Veeramachaneni. "MLFriend: Interactive Prediction Task Recommendation for Event-Driven Time-Series Data." In CoRR abs/1906.12348 (2019) .

Teaching Experience

- **UCF** — Orlando, Florida, US
  Assistant Professor, Department of CS — *August 2024 - Current*
  Courses Taught: Machine Learning (Graduate).

- **Auburn University** — Auburn, Alabama, US
  Assistant Professor, Department of CSSE — *Jan 2020 - April 2024*
  Courses Taught: Information Retrieval (Senior Undergrad), Natural Language Processing (Graduate), Machine Learning (Graduate/Undergrad), Artificial Intelligence (Graduate/Undergrad).

- **University of Illinois at Urbana-Champaign (UIUC)** — Urbana, Illinois, US
  Teaching Assistant, Department of Computer Science — *Fall 2017 and 2018*
  Courses Taught: Advanced Information Retrieval (Graduate), Text Mining Seminar (Graduate).

- **Bangladesh University of Engineering and Technology (BUET)** Dhaka, Bangladesh
  Lecturer, Department of Computer Science & Engineering *May 2012 - August 2014*
  Courses Taught: Computer Graphics, Computer Architecture, Operating Systems, etc (Undergrad).

Selected External Services

- **Action Editor**: Transactions of the Association for Computational Linguistics (TACL), [2025 - Current]
- **Communication Chair**: ACL Rolling Reviews (ARR) Initiative, [2024 - Current]
- **Action Editor**: ACL Rolling Reviews, [2021 - 2023]
- **Panelist**: NSF/CISE Grant Proposal Review Panel, [2026 RITEL, 2020, 2021]
- **Student Volunteer Committee Chair**: EMNLP 2024
- **Tutorial Track Chair**: CIKM, 2022
- **PC Member**: EMNLP, ACL, SIGIR, CIKM, WSDM, IUI (Multiple Years: 2017 - 2023)
- **Reviewer**: IEEE Transactions on Knowledge and Data Engineering (TKDE), Knowledge and Information Systems Journal, Neurocomputing Journal
- **PC Member**: WSDM Demo Track [2019, 2018], IUI Demo Track [2018, 2019]

Invited Talks

- **AI Expert Panelist on webinar at UCF, March 2026**: "Panel: Human–AI Collaboration: Insights from Interdisciplinary Research Teams."
- **Panelist on UCF Spring AI Research Day Panel discussion, February 2026**: "Panel: AI Education/Teaching"
- **Keynote Speaker at DAAD RISE program, Heidelberg University, Germany, July 2024**: "Being a Visiting Researcher in Germany."
- **Guest Speaker at University of Mannheim, Germany, July 2024**: "Democratizing AI through Controlled Narrative Generation and Conversational Alignment".
- **Guest Speaker at the University of Munich (LMU), Germany, June 2024**: "Democratizing AI through Controlled Narrative Generation and Conversational Alignment".
- **Guest Speaker at University of Würzburg, Germany, June 2024**: "Democratizing AI through Controlled Narrative Generation and Conversational Alignment".
- **Guest Speaker at Saarland University, Germany, June 2024**: "Democratizing AI through Controlled Narrative Generation and Conversational Alignment".
- **Guest Speaker at Technical University Darmstadt, Germany, June 2024**: "Democratizing AI through Controlled Narrative Generation and Conversational Alignment".
- **Invited Talk at University of Pittsburgh, January 2024**: "Democratizing AI through Controlled Narrative Generation and Knowledge Grounding".
- **Invited Talk at University of Buffalo, December 2023**: "Democratizing AI through Controlled Narrative Generation and Knowledge Grounding".
- **"Artificial Intelligence Initiative" Invited Speaker at University of Central Florida, November 2023**: "Democratizing AI through Controlled Narrative Generation and Knowledge Grounding".
- **Future Faculty Workshop Panelist, October 2023**: "Why I chose Academia?"
- **AI@AU Forum, September 2023**: "Democratizing AI with Information Assurance and Knowledge Grounding"
- **Apple Knowledge Platform Tech Talk, February 2023**: "Data Science for 'All' "
- **EMNLP 2022 Oral Presentation**: "SEM-F1: an Automatic Way for Semantic Evaluation of Multi-Narrative Overlap Summaries at Scale"
- **ACL 2022 Oral Presentation**: "Revisiting Automatic Evaluation of Extractive Summarization Task: Can We Do Better than ROUGE?"
- **COLING 2022 Oral Presentation**: "Semantic Overlap Summarization among Multiple Alternative Narratives: an Exploratory Study"
- **NSF EPSCoR Workshop: Artificial Intelligence (AI) with No-Boundary Thinking (NBT), April 2022**: "Data Science for 'All' "
- **Illinois Computer Science Speaker Series, Fall 2021**: "Data Science for 'All' "
- **CIKM 2020 Oral Presentation**: "Empirical Analysis of Impact of Query-Specific Customization of nDCG: A Case-Study with Learning-to-Rank Methods"
- **CIKM 2018 Oral Presentation**: "JIM: Joint Influence Modeling for Collective Search Behavior"
- **SIGIR 2017 Oral Presentation**: "On Application of Learning to Rank for E-Commerce Search"
- **CIKM 2016 Oral Presentation**: "Generative Feature Language Models for Mining Implicit Features from Customer Reviews"

Media Coverage and Expert Opinion

- **Auburn College of Engineering**: Assistant professor in CSSE earns two grants that combine for $1.25 million
- **AL.com**: Rise of the Chatbot: Alabama lawmakers confront questions about artificial intelligence
- **Auburn College of Engineering**: Artificial intelligence expert in CSSE weighs in on the rise of chatbots
- **Real World Data Science**: FOOD FOR THOUGHT: FIRST PLACE WINNERS – AUBURN BIG DATA

Awards Received by my Students

- **Auburn Outstanding Doctoral Student Award**: Naman Bansal [2023], Mousumi Akter [2023]
- **Auburn University Undergraduate Research Fellowship**: Hugh Williams [2023]
- **100+ Women Strong Outstanding Ph.D. Student**: Souvika Sarkar [2023], Mousumi Akter [2023]
- **Best poster at AU Engineering Research Showcase**: Naman Bansal [2021], Mousumi Akter [2022]

Current Student Advisees

- Ph.D. Students
    1. Alex "Ralph" Knipper [2020 - Current]
    2. Md. Mahadi Hasan "Sibat" [2021 - Current]
    3. Eftekhar Hossain [2024 - Current]
    4. Faria Binte Kader [2024 - Current]
    5. Adam Bawatneh [2025 - Current]

Alumni

- Ph.D. Alumni
    1. Yash Mahajan, Postdoc, UCF (Graduated 2025)
    2. Souvika Sarkar, Assistant Professor, Wichita State University (Graduated 2024)
    3. Naman Bansal, Software Engineer, GOOGLE (Graduated 2024)
    4. Mousumi Akter, Researcher, TU Dortmund (Graduated 2023)
    5. Dongji Feng, Assistant Professor, California State University, Monterey Bay (Graduated 2023)
- M.S. Alumni
    1. Hoang Bao Truong [2024 - 2026], UCF
    2. John Salvador [2022 - 2024], Auburn University
- B.S. Alumni
    1. Matthew Freestone, Fall 2024, Software Engineer, Reliable MicroSystems, Nashville, Tennessee.
    2. Hugh Williams, Spring 2025, M.S. at the University of Chicago.
    3. Saksham Goel, Fall 2021, Software Engineer, Microsoft.
- Past Visiting Ph.D. Students
    1. Sagar Sapkota [2024 - 2025], UCF
    2. Sanjeev Sinha [2022 - 2024], Auburn University
    3. Sri Ram Pavan Kumar Guttikonda [2023 - 2024], Auburn University

Internal Services at UCF

1. Ph.D. Dissertation Committee Member
   (a) Aakash Kumar
   (b) Fahimul Aleem
   (c) Yogesh Agrawal
2. Undergraduate COT 3960 Advising (Post Foundational Exam)
   (a) Jordan Rodriguez, 2026
   (b) Nicholas Calicchio, 2026
   (c) Nico Riley Burgos, 2025
   (d) Travis Burns, 2025
   (e) Cade Cochran, 2025
   (f) Alexis Cameron, 2025
   (g) Lianet Martin, 2025
   (h) Toricelli Carestil, 2025
   (i) Skyler Kuca, 2025
   (j) Nicholas Calicchio, 2025
   (k) Jeremiah Registre, 2025
   (l) Nikolai Colon, 2025
   (m) Griffin Cobb, 2024
   (n) Eli Suansing, 2024
   (o) Richard Phillips, 2024
   (p) Noah Fourari, 2024
   (q) David Little, 2024
3. EXCEL URE Faculty Mentor
   (a) Tanisha Gandhi, Spring 2026
   (b) Antoine Albrecht, Spring 2026
   (c) Camila Barbosa, Spring 2026
   (d) Jenna Curls, Spring 2026

Internal Services at Auburn

- Ph.D. Dissertation Committee Member.
  1. Gabrielle Taylor (Graduated Summer, 2023)
  2. Zijie Zhang (Graduated Summer, 2023)
  3. Bo Hui (Graduated Spring, 2023)
  4. Hung Nguyen (Graduated Spring, 2023)
  5. Chengfei Wang (Graduated Fall, 2022)
  6. Chao Jiang (Graduated Summer, 2022)
  7. Kenan Xiao (Graduated Summer, 2022)
  8. Ting Cao (Graduated Summer, 2021)
  9. Wenyu Zhu (Graduated Summer, 2021)
- Ph.D. Dissertation University Reader.
  1. Yuqiao Zhang (Graduated Summer, 2022)
  2. Ziqi Zhou (Graduated Summer, 2021)

- Departmental Service.
    1. CSSE Faculty Search Committee, 2022-23.
    2. CSSE Faculty Search Committee, 2020-21.
- College Level Service.
    1. College of Engineering Graduate Recruiting Committee Member (2023-24)
- University Level Service.
    1. AI@AU Initiative Education Committee.
    2. AI@AU Initiative Research and Equipment Committee Member (2022-24)
    3. AI@AU Initiative Presentation Schedule Organizing Committee Member (2022-24)